MeetMesh: Communication Between Personal AI Agents for Meeting Scheduling

CSE 476/575 Mobile Communication Networks – Term Project Proposal – Fall 2026

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1. Project Overview

- Today's Internet protocols are designed for people using applications. In the future, AI agents will communicate with each other on behalf of their users.
- Today, a group finds a meeting time by messaging back and forth or by sharing full calendars.
- Our goal: a mobile app in which each user's AI agent finds the other people's agents and agrees on a suitable time, without sharing the full calendar.

2. Project Objective

Redesign Internet services for AI agents in two ways: (1) redesign DNS at the application layer as an **Agent Directory** that finds agents by name or capability, and (2) add a new **Agent Communication Layer** that lets agents negotiate and exchange information reliably and securely over mobile networks. Meeting scheduling is used as the example application.

3. Scope of Work

- Agent Directory: a redesigned DNS for agents (application layer)
- The new Agent Communication Layer above TCP
- Personal AI agents that negotiate meeting times for one or more people
- A mobile app with profiles, friend search, meeting requests and a calendar view

4. Technical Approach

Redesigned application layer – Agent Directory. DNS only maps an exact host name to an IP address. Our directory maps a name or a capability (e.g., "study room booking") to an agent ID; an embedding model matches by meaning, results are cached with a short TTL, and users decide who can find them.

New layer – Agent Communication Layer. It sits between TCP and the application (Figure 1); the network, link and physical layers are not changed. Like SSL, it is a library on top of TCP.

- **Addressing:** Each agent has a fixed ID; a registrar maps it to its current IP address, like SIP.
- **Message protocol:** REQUEST, PROPOSE, ACCEPT, REJECT, CONFIRM and ACK in JSON; a group request goes to every participant.
- **Reliability:** Sequence numbers, ACKs, timeout and retransmission. Messages for offline agents wait in a queue at the registrar.
- **Security:** TLS, message authentication code and nonce; only free/busy data is shared.

AI component

- An LLM turns a sentence like "When can I meet Ahmet and Mehmet this week?" into a REQUEST.
- Each agent reads free/busy times from Google Calendar and ranks free slots by preference.
- The requesting agent picks the common slot with the highest total score. The slot is held until everyone confirms, then it is added to all calendars.

5. Success Criteria

The project is successful if the Agent Directory finds the correct agent and a group agrees on a meeting time without sharing full calendars, even when messages are lost or delayed, or some agents are offline.

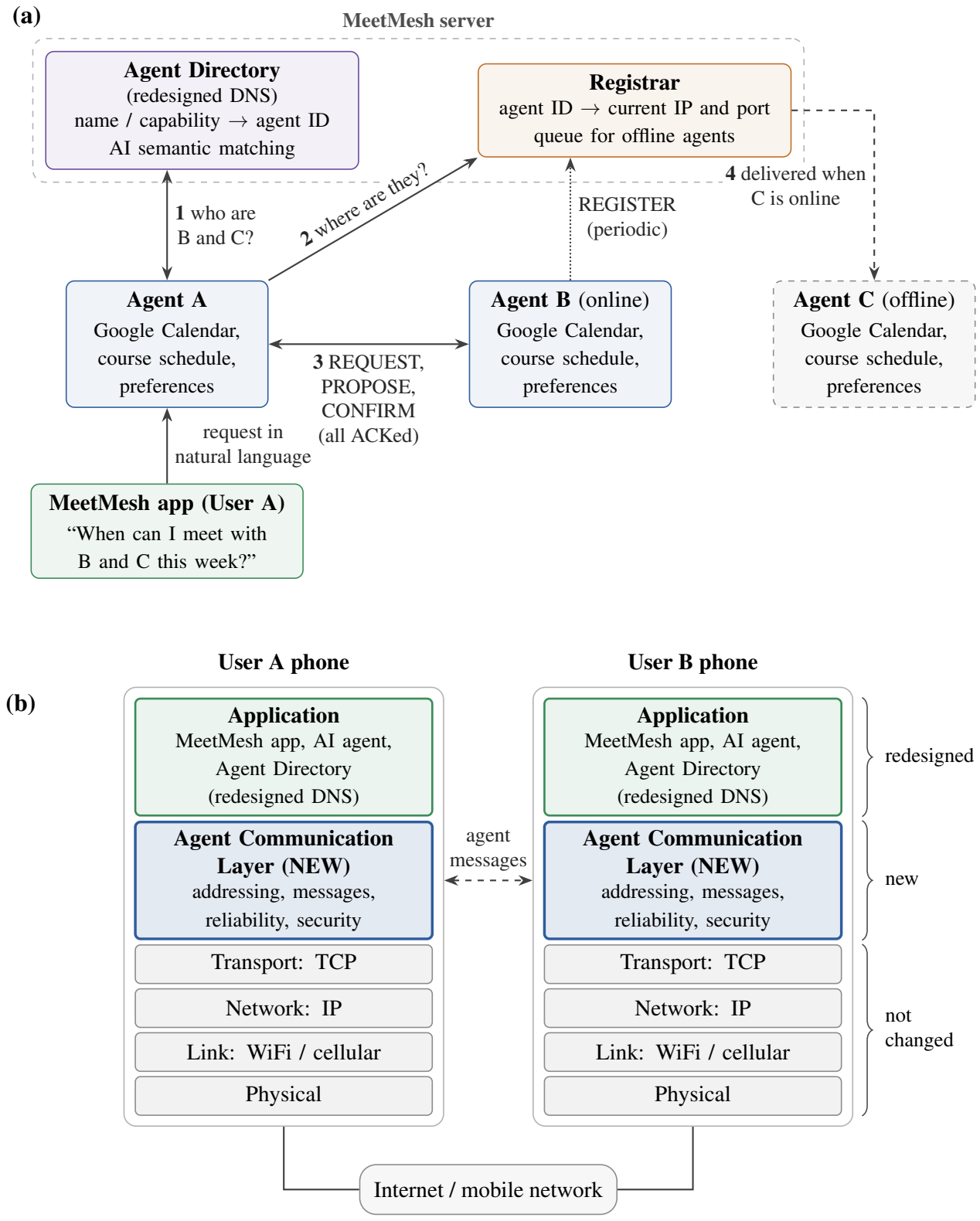

Figure 1: System architecture of MeetMesh. (a) Agent A asks the Agent Directory who B and C are (1) and the registrar where they are (2), then negotiates a meeting time with them (3); a message for an offline agent is kept in the registrar's queue and delivered when the agent comes back online (4). (b) The application layer is redesigned (Agent Directory), a new Agent Communication Layer is added between the application and TCP, and the lower layers are not changed.